



MAP 112 - Álgebra Linear

Matrizes Inversas

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1. Para cada matriz

- verifique se é invertível;
- se for, calcule a inversa;
- confira multiplicando.

ex.  $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$   $(1 \cdot 5) - (2 \cdot 3)$   
 $= 5 - 6$   
 $= -1 \rightarrow$  Matriz Invertível

$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

$L2 \rightarrow L2 - 3L1$   $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{bmatrix}$   
 $= 3 - 3 \cdot 1 = 0$   
 $= 5 - 3 \cdot 2 = -1$   
 $= 0 - 3 \cdot 1 = -3$   
 $= 1 - 3 \cdot 0 = 1$

$L2 \rightarrow (-1)L2$   $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{bmatrix}$   
 $= -1 \cdot 1 = -1$   
 $= -1 \cdot 3 = -3$   
 $= -1 \cdot (-1) = 1$

$$L1 \rightarrow L1 - 2L2$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 3 = -5$$

$$= 0 - 2 \cdot (-1) = 2$$

$$\begin{bmatrix} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot (-5) + 2 \cdot 3 & 1 \cdot 2 + 2 \cdot (-1) \\ 3 \cdot (-5) + 5 \cdot 3 & 3 \cdot 2 + 5 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -5 + 6 & 2 - 2 \\ -15 + 15 & 6 - 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det. \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = (2 \cdot 1) - (1 \cdot 1) = 2 - 1 = 1$$

$= 1 \rightarrow$  Matriz Invertível

$$\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$\text{Trocar } L1 \text{ por } L2 \quad \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{bmatrix}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 0 - 2 \cdot 1 = -2$$

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{bmatrix}$$

$$|2 \rightarrow -1|$$

$$= -1 \cdot -1 = 1$$

$$= -1 \cdot 1 = -1$$

$$= -1 \cdot -2 = 2$$

$$\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ -1 & 2 \end{vmatrix}$$

$$|1 \rightarrow 1|$$

$$= 1 - 1 = 0$$

$$= 0 - (-1) = 1$$

$$= 1 - 2 = -1$$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix}$$

$$\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} = \begin{vmatrix} 2 \cdot 1 + 1 \cdot (-1) & 2 \cdot (-1) + 1 \cdot 2 \\ 1 \cdot 1 + 1 \cdot (-1) & 1 \cdot (-1) + 1 \cdot 2 \end{vmatrix} = \begin{vmatrix} 2 - 1 & -2 + 2 \\ 1 - 1 & -1 + 2 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$A^{-1} = \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix}$$

$$\text{ic.} \begin{vmatrix} 3 & 6 \\ 1 & 2 \end{vmatrix} \begin{vmatrix} 3 \cdot 2 - 6 \cdot 1 \\ 1 \cdot 2 - 3 \cdot 6 \end{vmatrix} = \begin{vmatrix} 6 - 6 \\ 2 - 18 \end{vmatrix} = \begin{vmatrix} 0 \\ -16 \end{vmatrix}$$

Como  $\det A = 0$ , a matriz é não invertível

$$\text{id.} \begin{vmatrix} 4 & 9 \\ 2 & 6 \end{vmatrix} \begin{vmatrix} 4 \cdot 6 - 9 \cdot 2 \\ 2 \cdot 4 - 9 \cdot 6 \end{vmatrix} = \begin{vmatrix} 24 - 18 \\ 8 - 54 \end{vmatrix} = \begin{vmatrix} 6 \\ -46 \end{vmatrix}$$

$$A_{11} = (-1)^{1+1} \cdot 6 = (-1)^2 \cdot 6$$

$$= 1 \cdot 6 = 6$$

$$A_{12} = (-1)^{1+2} = (-1)^3 \cdot 2$$

$$= -1 \cdot 2 = -2$$

$$A_{21} = (-1)^{2+1} = (-1)^3 \cdot 9$$

$$= -1 \cdot 9 = -9$$

$$A_{22} = (-1)^{2+2} = (-1)^4 \cdot 4$$

$$= 1 \cdot 4 = 4$$

Stamper Matrix

$$\begin{bmatrix} 6 & -9 \\ -2 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -9 \\ -2 & 4 \end{pmatrix} = \begin{bmatrix} \frac{6}{10} & -\frac{9}{10} \\ -\frac{2}{10} & \frac{4}{10} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 9 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} \frac{6}{10} & -\frac{9}{10} \\ -\frac{2}{10} & \frac{4}{10} \end{bmatrix} = \begin{bmatrix} 4 \cdot \frac{6}{10} + 9 \cdot -\frac{2}{10} & 4 \cdot -\frac{9}{10} + 9 \cdot \frac{4}{10} \\ 2 \cdot \frac{6}{10} + 6 \cdot -\frac{2}{10} & 2 \cdot -\frac{9}{10} + 6 \cdot \frac{4}{10} \end{bmatrix} = \begin{bmatrix} \frac{24}{10} - \frac{18}{10} & -\frac{36}{10} + \frac{36}{10} \\ \frac{12}{10} - \frac{12}{10} & -\frac{18}{10} + \frac{24}{10} \end{bmatrix} = \begin{bmatrix} \frac{6}{10} & 0 \\ 0 & \frac{6}{10} \end{bmatrix}$$

$$A^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -9 \\ -2 & 4 \end{pmatrix} \text{ oder } \begin{bmatrix} \frac{6}{10} & -\frac{9}{10} \\ -\frac{2}{10} & \frac{4}{10} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix} \quad 3 \cdot 1 - 2 \cdot (-1)$$

$$= 3 + 2$$

$$= 5$$

$$A_{11} = (-1)^{1+1} \cdot 1 = -1^2 \cdot 1$$

$$= 1 \cdot 1 = 1$$

$$A_{12} = (-1)^{1+2} \cdot 2 = -1^3 \cdot 2$$

$$= -1 \cdot 2 = -2$$

$$A_{21} = (-1)^{2+1} \cdot (-1) = -1^3 \cdot (-1)$$

$$= -1 \cdot (-1) = 1$$

$$A_{22} = (-1)^{2+2} \cdot 3 = -1^4 \cdot 3$$

$$= 1 \cdot 3 = 3$$

Inverser Matrix

$$\begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} 3 \cdot \frac{1}{5} + (-1) \cdot (-\frac{2}{5}) & 3 \cdot \frac{1}{5} + (-1) \cdot \frac{3}{5} \\ 2 \cdot \frac{1}{5} + 1 \cdot (-\frac{2}{5}) & 2 \cdot \frac{1}{5} + 1 \cdot \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{3}{5} + \frac{2}{5} & \frac{3}{5} - \frac{3}{5} \\ \frac{2}{5} - \frac{2}{5} & \frac{2}{5} + \frac{3}{5} \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ -2 & 3 \end{pmatrix} \text{ oder } \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{5}{5} & \frac{0}{5} \\ \frac{0}{5} & \frac{5}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} \quad 1 \cdot 2 - 0 \cdot 5$$

$$= 2$$

$$A_{11} = (-1)^{1+1} \cdot 2 = (-1)^0 \cdot 2$$

$$= 1 \cdot 2 = 2$$

$$A_{12} = (-1)^{1+2} \cdot 5 = (-1)^3 \cdot 5$$

$$= -1 \cdot 5 = -5$$

$$A_{21} = (-1)^{2+1} \cdot 0 = (-1)^3 \cdot 0$$

$$= -1 \cdot 0 = 0$$

$$A_{22} = (-1)^{2+2} \cdot 1 = (-1)^4 \cdot 1$$

$$= 1 \cdot 1 = 1$$

$$\begin{bmatrix} 2 & -5 \\ 0 & 1 \end{bmatrix}$$

Transfer Matrix

$$\begin{bmatrix} 2 & 0 \\ -5 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ -5 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{5}{2} & \frac{1}{2} \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -\frac{5}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot (-\frac{5}{2}) & 1 \cdot 0 + 0 \cdot \frac{1}{2} \\ 5 \cdot 1 + 2 \cdot (-\frac{5}{2}) & 5 \cdot 0 + 2 \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 - 0 & 0 + 0 \\ 5 - 5 & 0 + 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ -5 & 1 \end{pmatrix} \text{ oder } \begin{bmatrix} 1 & 0 \\ -\frac{5}{2} & \frac{1}{2} \end{bmatrix}$$

$$\text{eg. } \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \quad \begin{array}{l} 2 \cdot 4 - 3 \cdot 1 \\ = 8 - 3 \\ = 5 \end{array}$$

$$\begin{bmatrix} 2 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \quad \text{Swan L1 per L2} \quad \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 0 \end{bmatrix}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 3 - 2 \cdot 2 = -5$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 0 - 2 \cdot 1 = -2$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -5 & 1 & -2 \end{bmatrix}$$

$$L3 \rightarrow -1L2$$

$$5$$

$$= -\frac{1}{5} \cdot (-5) = \frac{5}{5} = 1$$

$$= -\frac{1}{5} \cdot 1 = -\frac{1}{5}$$

$$= -\frac{1}{5} \cdot (-2) = \frac{2}{5}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$L1 \rightarrow L1 - 2L2$$

$$2 - 2 \cdot 1 = 0$$

$$0 - 2 \cdot (-\frac{1}{5}) = \frac{2}{5}$$

$$1 - 2 \cdot \frac{2}{5} = 1 - \frac{4}{5} = \frac{1}{5}$$

$$\begin{bmatrix} 1 & 0 & \frac{2}{5} & -\frac{3}{5} \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \quad \begin{array}{l} \frac{4}{5} - \frac{3}{5} \\ -\frac{1}{5} + \frac{2}{5} \end{array} = \begin{array}{l} 2 \cdot \frac{4}{5} + 3 \cdot (-\frac{1}{5}) \\ 1 \cdot \frac{4}{5} + 2 \cdot (-\frac{1}{5}) \end{array} \quad \begin{array}{l} 2 \cdot (-\frac{3}{5}) + 3 \cdot \frac{2}{5} \\ -1 \cdot (-\frac{3}{5}) + 2 \cdot \frac{2}{5} \end{array} = \begin{array}{l} \frac{8}{5} - \frac{3}{5} \\ \frac{4}{5} - \frac{2}{5} \end{array} \quad \begin{array}{l} -\frac{6}{5} + \frac{6}{5} \\ -\frac{3}{5} + \frac{4}{5} \end{array}$$

$$\begin{bmatrix} \frac{5}{5} & 0 \\ 0 & \frac{5}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$\text{ii. } \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad (1 \cdot 4) - (2 \cdot 2) = 4 - 4 = 0$$

Como  $\det A = 0$ , a matriz é não invertível.

$$\text{ii. } \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix} \quad (3 \cdot 2) - (6 \cdot 1) = 6 - 6 = 0$$

Como  $\det A = 0$ , a matriz é não invertível.

$$\text{iii. } \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad (1+1+0) - (0+0+0) = 2 - 0 = 2$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= 0 - 1 = -1$$

$$= 1 - 0 = 1$$

$$= 0 - 1 = -1$$

$$= 0 - 0 = 0$$

$$= 1 - 0 = 1$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{bmatrix}$$

$$L3 \rightarrow L3 + L2$$

$$= -1 + 1 = 0$$

$$= 1 + 1 = 2$$

$$= -1 + 0 = -1$$

$$= 0 + 1 = 1$$

$$= 1 + 0 = 1$$

$$\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array}$$

$$L4 \rightarrow L4 - L2$$

$$= 1 - 1 = 0$$

$$= 0 - 1 = -1$$

$$= 1 - 0 = 1$$

$$= 0 - 1 = -1$$

$$= 0 - 0 = 0$$

$$\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array}$$

$$L3 \rightarrow 4 \cdot L3$$

$$= 2$$

$$= \frac{1}{2} \cdot 2 = 0$$

$$= \frac{1}{2} \cdot (-1) = -\frac{1}{2}$$

$$= \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$$= \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$$\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}$$

$$L2 \rightarrow L2 - L3$$

$$= 1 - 1 = 0$$

$$= 0 - \frac{1}{2} = -\frac{1}{2}$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$$= 0 - \frac{1}{2} = -\frac{1}{2}$$

$$\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}$$

$$L_1 - D L_1 + L_3$$

$$= -1 + 1 = 0$$

$$= 1 + (-\frac{1}{2}) = \frac{1}{2}$$

$$= -1 + \frac{1}{2} = -\frac{1}{2}$$

$$= 0 + \frac{1}{2} = \frac{1}{2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} =$$

$$= \begin{bmatrix} 1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} + 0 \cdot -\frac{1}{2} & 1 \cdot -\frac{1}{2} + 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} & 1 \cdot \frac{1}{2} + 1 \cdot -\frac{1}{2} + 0 \cdot \frac{1}{2} \\ 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} + 1 \cdot -\frac{1}{2} & 0 \cdot -\frac{1}{2} + 1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} & 0 \cdot \frac{1}{2} + 1 \cdot -\frac{1}{2} + 1 \cdot \frac{1}{2} \\ 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} + 1 \cdot -\frac{1}{2} & 1 \cdot -\frac{1}{2} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} & 1 \cdot \frac{1}{2} + 0 \cdot -\frac{1}{2} + 1 \cdot \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} + \frac{1}{2} - 0 & -\frac{1}{2} + \frac{1}{2} + 0 & \frac{1}{2} - \frac{1}{2} + 0 \\ 0 + \frac{1}{2} - \frac{1}{2} & -0 + \frac{1}{2} + \frac{1}{2} & 0 - \frac{1}{2} + \frac{1}{2} \\ \frac{1}{2} + 0 - \frac{1}{2} & -\frac{1}{2} + 0 + \frac{1}{2} & \frac{1}{2} - 0 + \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\text{Ans. } \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = (2+0+1) - (0+0+0) = 3$$

$$\begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$



|   |   |   |   |   |   |                     |   |   |   |   |   |   |
|---|---|---|---|---|---|---------------------|---|---|---|---|---|---|
| 2 | 0 | 1 | 1 | 0 | 0 |                     | 1 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | Subtraher L1 per L2 | 2 | 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 |                     | 0 | 1 | 1 | 0 | 0 | 1 |

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 0 - 2 \cdot 1 = -2$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 0 - 2 \cdot 1 = -2$$

$$= 0 - 2 \cdot 0 = 0$$

|   |    |   |   |    |   |
|---|----|---|---|----|---|
| 1 | 1  | 0 | 0 | 1  | 0 |
| 0 | -2 | 1 | 1 | -2 | 0 |
| 0 | 1  | 1 | 0 | 0  | 1 |

$$\text{Subtraher L2 per L3}$$

|   |    |   |   |    |   |
|---|----|---|---|----|---|
| 1 | 1  | 0 | 0 | 1  | 0 |
| 0 | 1  | 1 | 0 | 0  | 1 |
| 0 | -2 | 1 | 1 | -2 | 0 |

$$L1 \rightarrow L1 - L2$$

$$= 1 - 1 = 0$$

$$= 0 - 1 = -1$$

$$= 0 - 0 = 0$$

$$= 1 - 0 = 1$$

$$= 0 - 1 = -1$$

|   |    |    |   |    |    |
|---|----|----|---|----|----|
| 1 | 0  | -1 | 0 | 1  | -1 |
| 0 | 1  | 1  | 0 | 0  | 1  |
| 0 | -2 | 1  | 1 | -2 | 0  |

$$L3 \rightarrow L3 + 2L2$$

$$= -2 + 2 \cdot 1 = 0$$

$$= 1 + 2 \cdot 1 = 3$$

$$= 1 + 2 \cdot 0 = 1$$

$$= -2 + 2 \cdot 0 = -2$$

$$= 0 + 2 \cdot 1 = 2$$

|   |   |    |   |    |    |
|---|---|----|---|----|----|
| 1 | 0 | -1 | 0 | 1  | -1 |
| 0 | 1 | 1  | 0 | 0  | 1  |
| 0 | 0 | 3  | 1 | -2 | 2  |

$$L3 \rightarrow L1 \cdot L3$$

$$3$$

$$= \frac{1}{3} \cdot 3 = 1$$

$$= \frac{1}{3} \cdot 4 = \frac{4}{3}$$

$$= \frac{1}{3} \cdot 2 = \frac{2}{3}$$

$$= \frac{1}{3} \cdot 2 = \frac{2}{3}$$

|   |   |    |               |                |               |
|---|---|----|---------------|----------------|---------------|
| 1 | 0 | -1 | 0             | 1              | -1            |
| 0 | 1 | 1  | 0             | 0              | 1             |
| 0 | 0 | 1  | $\frac{1}{3}$ | $-\frac{2}{3}$ | $\frac{2}{3}$ |

$$L4 \rightarrow L4 + L3$$

$$= -1 + 1 = 0$$

$$= 0 + \frac{1}{3} = \frac{1}{3}$$

$$= 1 + \frac{2}{3} = \frac{5}{3}$$

$$= -1 + \frac{2}{3} = -\frac{1}{3}$$

|   |   |   |               |                |                |
|---|---|---|---------------|----------------|----------------|
| 1 | 0 | 0 | $\frac{1}{3}$ | $\frac{1}{3}$  | $-\frac{1}{3}$ |
| 0 | 1 | 1 | 0             | 0              | 1              |
| 0 | 0 | 1 | $\frac{1}{3}$ | $-\frac{2}{3}$ | $\frac{2}{3}$  |

$$L2 \rightarrow L2 - L3$$

$$= 1 - 1 = 0$$

$$= 0 - \frac{1}{3} = -\frac{1}{3}$$

$$= 0 - (-\frac{2}{3}) = \frac{2}{3}$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

|   |   |   |                |                |                |
|---|---|---|----------------|----------------|----------------|
| 1 | 0 | 0 | $\frac{1}{3}$  | $\frac{1}{3}$  | $-\frac{1}{3}$ |
| 0 | 1 | 0 | $-\frac{1}{3}$ | $\frac{2}{3}$  | $\frac{1}{3}$  |
| 0 | 0 | 1 | $\frac{1}{3}$  | $-\frac{2}{3}$ | $\frac{2}{3}$  |

|   |   |   |                |                |                |
|---|---|---|----------------|----------------|----------------|
| 2 | 0 | 1 | $\frac{1}{3}$  | $\frac{1}{3}$  | $-\frac{1}{3}$ |
| 4 | 1 | 0 | $-\frac{1}{3}$ | $\frac{2}{3}$  | $\frac{1}{3}$  |
| 0 | 1 | 1 | $\frac{1}{3}$  | $-\frac{2}{3}$ | $\frac{2}{3}$  |

|                                                                   |                                                                   |                                                                   |
|-------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|
| $2 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}$ | $2 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} + 1 \cdot \frac{2}{3}$ | $2 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3}$ |
| $1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}$ | $1 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} + 0 \cdot \frac{2}{3}$ | $1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3}$ |
| $0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}$ | $0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} + 1 \cdot \frac{2}{3}$ | $0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3}$ |

|                                           |                                           |                                            |               |               |               |
|-------------------------------------------|-------------------------------------------|--------------------------------------------|---------------|---------------|---------------|
| $\frac{2}{3} - \frac{0}{3} + \frac{1}{3}$ | $\frac{2}{3} + \frac{0}{3} - \frac{2}{3}$ | $-\frac{2}{3} + \frac{0}{3} + \frac{2}{3}$ | $\frac{3}{3}$ | $\frac{0}{3}$ | $\frac{0}{3}$ |
| $\frac{1}{3} - \frac{1}{3} + \frac{0}{3}$ | $\frac{1}{3} + \frac{2}{3} - \frac{0}{3}$ | $-\frac{1}{3} + \frac{1}{3} + \frac{0}{3}$ | $\frac{0}{3}$ | $\frac{3}{3}$ | $\frac{0}{3}$ |
| $\frac{0}{3} - \frac{1}{3} + \frac{1}{3}$ | $\frac{0}{3} + \frac{2}{3} - \frac{2}{3}$ | $-\frac{0}{3} + \frac{1}{3} + \frac{2}{3}$ | $\frac{0}{3}$ | $\frac{0}{3}$ | $\frac{3}{3}$ |

tilibra

|   |   |   |
|---|---|---|
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

$$A^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{vmatrix} = 1(4+12+6) - (12+6+4) = 22 - 22 = 0$$

Como  $\det A = 0$ , a matriz é não invertível.

$$\text{vm} = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 2 & 3 & 4 \end{vmatrix} = 1(4+0+0) - (2+0+0) = 4 - 2 = 2$$

|   |   |   |
|---|---|---|
| 1 | 2 | 1 |
| 0 | 1 | 0 |
| 2 | 3 | 4 |

$$L3 \rightarrow L3 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 3 - 2 \cdot 2 = -1$$

$$= 4 - 2 \cdot 1 = 2$$

$$= 0 - 2 \cdot 1 = -2$$

$$= 0 - 2 \cdot 0 = 0$$

$$= 1 - 2 \cdot 0 = 1$$

|   |    |   |
|---|----|---|
| 1 | 2  | 1 |
| 0 | 1  | 0 |
| 0 | -1 | 2 |

$$L3 \rightarrow L3 + L2$$

$$= -1 + 1 = 0$$

$$= 2 + 0 = 2$$

$$= -2 + 0 = -2$$

$$= 0 + 1 = 1$$

$$= 1 + 0 = 1$$

|   |   |   |    |   |   |   |
|---|---|---|----|---|---|---|
| 1 | 2 | 1 | 1  | 0 | 1 | 0 |
| 0 | 1 | 0 | 0  | 1 | 0 | 0 |
| 0 | 0 | 2 | -2 | 1 | 1 | - |

$$L1 \rightarrow L1 - 2L2$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 0 - 2 \cdot 1 = -2$$

$$= 0 - 2 \cdot 0 = 0$$

|   |   |   |    |   |    |   |
|---|---|---|----|---|----|---|
| 1 | 0 | 1 | 1  | 0 | -2 | 0 |
| 0 | 1 | 0 | 0  | 1 | 0  | 0 |
| 0 | 0 | 2 | -2 | 1 | 1  | - |

$$L3 \rightarrow L3 - L1$$

$$= 2 - 1 = 1$$

$$= (-2) - 1 = -3$$

$$= -1 - (-2) = 1$$

$$= 1 - 0 = 1$$

|   |   |   |   |    |   |   |
|---|---|---|---|----|---|---|
| 1 | 0 | 1 | 1 | -2 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1  | 0 | 0 |
| 0 | 0 | 1 | 2 | -3 | 1 | - |

$$L1 \rightarrow L1 - L3$$

$$= 1 - 1 = 0$$

$$= 1 - (-3) = -4$$

$$= -2 - (-3) = 1$$

$$= 0 - (1) = -1$$

|   |   |   |    |    |    |   |
|---|---|---|----|----|----|---|
| 1 | 0 | 0 | -4 | 5  | 1  | - |
| 0 | 1 | 0 | 0  | 1  | 0  | 0 |
| 0 | 0 | 1 | 2  | -3 | -1 | - |

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 0 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} -4 & 5 & 1 \\ 0 & 1 & 0 \\ 2 & -3 & -1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 \cdot (-4) + 1 \cdot 1 + 2 \cdot 2 & 1 \cdot 5 + 2 \cdot 1 + 1 \cdot (-3) & 1 \cdot 1 + 2 \cdot 0 + 1 \cdot (-1) \\ 0 \cdot (-4) + 1 \cdot 0 + 0 \cdot 2 & 0 \cdot 5 + 1 \cdot 1 + 0 \cdot (-3) & 0 \cdot 1 + 1 \cdot 0 + 0 \cdot (-1) \\ 2 \cdot (-4) + 3 \cdot 0 + 4 \cdot 2 & 2 \cdot 5 + 3 \cdot 1 + 4 \cdot (-3) & 2 \cdot 1 + 3 \cdot 0 + 4 \cdot (-1) \end{bmatrix}$$

$$\begin{bmatrix} -4 + 1 + 4 & 5 - 2 - 3 & 1 + 0 - 1 \\ 0 + 0 + 0 & 0 + 1 + 0 & 0 + 0 + 0 \\ -8 + 0 + 8 & 10 + 3 - 12 & 2 - 0 - 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -4 & 5 & 1 \\ 0 & 1 & 0 \\ 2 & -3 & -1 \end{bmatrix}$$

ex.  $\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$   $\begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$   $(3+1) - (2+2) = 2 - 4 = -2$   
 $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$   $\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$   $2 - 1 = 1$   
 $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$   $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$   $1 - 0 = 1$

Como  $\det A = 0$ , a matriz é não invertível

ex.  $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$   $\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}$   $(1+4) - (3+0) = 5 - 3 = 2$   
 $\begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$   $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$   $2 - 1 = 1$   
 $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$   $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$   $1 - 0 = 1$

$$A_{11} = (-1)^{1+1} \cdot 1 = -1^9 \cdot 1$$

$$\text{Minor} = (1 \cdot 1) - (0 \cdot 2)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= -1 \cdot 1 = -1$$

$$= 1 - 0$$

$$= 1 - 0$$

$$= 1$$

$$= 1$$

$$A_{12} = (-1)^{1+2} \cdot 2 = -1^3 \cdot 2$$

$$\text{Minor} = (2 \cdot 1) - (0 \cdot 3)$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= -1 \cdot 2 = -2$$

$$= 2 - 0$$

$$= 2 - 0$$

$$= 2$$

$$= 2$$

$$A_{13} = (-1)^{1+3} \cdot 1 = -1^4 \cdot 1$$

$$\text{Minor} = (2 \cdot 2) - (1 \cdot 3)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= 1 \cdot 1 = 1$$

$$= 4 - 3$$

$$= 1 - 0$$

$$= 1$$

$$= 1$$

$$A_{21} = (-1)^{2+1} \cdot 1 = -1^3 \cdot (-2)$$

$$\text{Minor} = (0 \cdot 1) - (1 \cdot 2)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= -1 \cdot (-2) = 2$$

$$= 0 - 2$$

$$= 0 - 2$$

$$= -2$$

$$= -2$$

$$A_{22} = (-1)^{2+2} \cdot (-2) = -1^4 \cdot (-2)$$

$$\text{Minor} = (1 \cdot 1) - (1 \cdot 3)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= 1 \cdot (-2) = -2$$

$$= 1 - 3$$

$$= 1 - 3$$

$$= -2$$

$$= -2$$

$$A_{23} = (-1)^{2+3} \cdot 2 = -1^5 \cdot 2$$

$$\text{Minor} = (1 \cdot 2) - (0 \cdot 3)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= -1 \cdot 2 = -2$$

$$= 2 - 0$$

$$= 2 - 0$$

$$= 2$$

$$= 2$$

$$A_{31} = (-1)^{3+1} \cdot 1 = -1^4 \cdot 1$$

$$\text{Minor} = (0 \cdot 0) - (1 \cdot 1)$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= -1 \cdot 1 = -1$$

$$= 1 \cdot 1$$

$$= 1 \cdot 1$$

$$= 1$$

$$= 1$$

$$A_{32} = (-1)^{3+2} \cdot (-2) = -1^5 \cdot (-1)$$

$$= -1 \cdot (-2) = 2$$

$$\text{Minor} = (1 \cdot 0) - (1 \cdot 2)$$

$$= 0 - 2$$

$$= -2$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

$$A_{33} = (-1)^{3+3} \cdot 1 = -1^6 \cdot 1$$

$$= 1 \cdot 1 = 1$$

$$\text{Minor} = (1 \cdot 1) - (0 \cdot 2)$$

$$= 1 - 0$$

$$= 1$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix}$$

|    |    |    |
|----|----|----|
| 1  | -2 | 1  |
| 2  | -2 | -2 |
| -1 | 2  | 1  |

Transpose Matrix

|    |    |    |
|----|----|----|
| 1  | 2  | -1 |
| -2 | -2 | 2  |
| 1  | -2 | 1  |

$$A^{-1} = \frac{1}{|A|} \cdot \begin{vmatrix} 1 & 2 & 1 \\ 2 & -2 & 2 \\ 1 & -2 & 1 \end{vmatrix} = \frac{1}{-2} \cdot \begin{vmatrix} 1 & 2 & 1 \\ 2 & -2 & 2 \\ 1 & -2 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} \cdot \begin{vmatrix} \frac{1}{2} & 1 & -\frac{1}{2} \\ -1 & -1 & 1 \\ \frac{1}{2} & -1 & \frac{1}{2} \end{vmatrix} =$$

$$\begin{aligned} 1 \cdot \frac{1}{2} + 0 \cdot (-1) + 1 \cdot \frac{1}{2} &= 1 \cdot 1 + 0 \cdot (-1) + 1 \cdot (-1) & 1 \cdot -\frac{1}{2} + 0 \cdot 1 + 1 \cdot \frac{1}{2} \\ 2 \cdot \frac{1}{2} + 1 \cdot (-1) + 0 \cdot \frac{1}{2} &= 2 \cdot 1 + 1 \cdot (-1) + 0 \cdot (-1) & 2 \cdot -\frac{1}{2} + 1 \cdot 1 + 0 \cdot (-\frac{1}{2}) \\ 3 \cdot \frac{1}{2} + 2 \cdot (-1) + 1 \cdot \frac{1}{2} &= 3 \cdot 1 + 2 \cdot (-1) + 1 \cdot (-1) & 3 \cdot -\frac{1}{2} + 2 \cdot 1 + 1 \cdot (\frac{1}{2}) \end{aligned}$$

|                                 |             |                                  |     |     |     |
|---------------------------------|-------------|----------------------------------|-----|-----|-----|
| $\frac{1}{2} + \frac{1}{2}$     | $1 - 1$     | $-\frac{1}{2} + \frac{1}{2}$     | $1$ | $0$ | $0$ |
| $\frac{2}{4} - 1$               | $2 - 1$     | $-1 + 1$                         | $0$ | $1$ | $0$ |
| $\frac{3}{8} - 2 + \frac{1}{2}$ | $3 - 2 - 1$ | $-\frac{3}{2} + 2 + \frac{1}{2}$ | $0$ | $0$ | $1$ |

$$A^{-1} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix} \text{ ou } \begin{bmatrix} \frac{1}{2} & 1 & -\frac{1}{2} \\ -1 & -1 & 1 \\ \frac{1}{2} & -2 & -1 \end{bmatrix}$$

1p.

|    |    |    |   |   |
|----|----|----|---|---|
| 11 | 12 | 13 | 2 | 1 |
| 2  | 1  | 0  | 2 | 1 |
| 31 | 32 | 33 | 1 | 3 |
| 1  | 3  | 1  | 0 | 0 |
| 31 | 32 | 33 | 0 | 0 |
| 0  | 1  | 2  | 0 | 0 |

$42 - (2+2) = 12 - 4 = 8$

$$A_{11} = (-1)^{1+1} \cdot 5 = -1^2 \cdot 5 = 1 \cdot 5 = 5$$

$$\text{Memor} = (3 \cdot 2) - (1 \cdot 1) = 6 - 1 = 5$$

|   |   |   |
|---|---|---|
| 2 | 1 | 0 |
| 1 | 3 | 1 |
| 0 | 1 | 2 |

$$A_{12} = (-1)^{1+2} \cdot 2 = -1^3 \cdot 2 = -1 \cdot 2 = -2$$

$$\text{Memor} = (1 \cdot 2) - (1 \cdot 0) = 2 - 0 = 2$$

|   |   |   |
|---|---|---|
| 2 | 1 | 0 |
| 1 | 3 | 1 |
| 0 | 1 | 2 |

$$A_{13} = (-1)^{1+3} \cdot 1 = -1^4 \cdot 1 = 1 \cdot 1 = 1$$

$$\text{Memor} = (1 \cdot 1) - (3 \cdot 0) = 1 - 0 = 1$$

|   |   |   |
|---|---|---|
| 2 | 1 | 0 |
| 1 | 3 | 1 |
| 0 | 1 | 2 |

$$A_{21} = (-1)^{2+1} \cdot 2 = -1^3 \cdot 2 = -1 \cdot 2 = -2$$

$$\text{Memor} = (1 \cdot 2) - (0 \cdot 1) = 2 - 0 = 2$$

|   |   |   |
|---|---|---|
| 2 | 1 | 0 |
| 1 | 3 | 1 |
| 0 | 1 | 2 |

$$A_{22} = (-1)^{3+2} \cdot 4 = -1^4 \cdot 4$$

$$= 1 \cdot 4 = 4$$

$$M_{22} = (2 \cdot 2) - (0 \cdot 0)$$

$$= 4 - 0$$

$$= 4$$

$$2 \quad 1 \quad 0$$

$$1 \quad 3 \quad 1$$

$$0 \quad 1 \quad 2$$

$$A_{23} = (-1)^{3+3} \cdot 2 = (-1)^5 \cdot 2$$

$$= -1 \cdot 2 = -2$$

$$M_{23} = (2 \cdot 1) - (1 \cdot 0)$$

$$= 2 - 0$$

$$= 2$$

$$2 \quad 1 \quad 0$$

$$1 \quad 3 \quad 1$$

$$0 \quad 1 \quad 2$$

$$A_{31} = (-1)^{3+1} \cdot 1 = (-1)^4 \cdot 1$$

$$= 1 \cdot 1 = 1$$

$$M_{31} = (1 \cdot 1) - (0 \cdot 3)$$

$$= 1 - 0$$

$$= 1$$

$$2 \quad 1 \quad 0$$

$$1 \quad 3 \quad 1$$

$$0 \quad 1 \quad 2$$

$$A_{32} = (-1)^{3+2} \cdot 2 = (-1)^5 \cdot 2$$

$$= -1 \cdot 2 = -2$$

$$M_{32} = (2 \cdot 1) - (0 \cdot 1)$$

$$= 2 - 0$$

$$= 2$$

$$2 \quad 1 \quad 0$$

$$1 \quad 3 \quad 1$$

$$0 \quad 1 \quad 2$$

$$A_{33} = (-1)^{3+3} \cdot 5 = (-1)^6 \cdot 5$$

$$= 1 \cdot 5 = 5$$

$$M_{33} = (2 \cdot 3) - (1 \cdot 1)$$

$$= 6 - 1$$

$$= 5$$

$$2 \quad 1 \quad 0$$

$$1 \quad 3 \quad 1$$

$$0 \quad 1 \quad 2$$

|    |    |    |
|----|----|----|
| 5  | -2 | 1  |
| -2 | 4  | -2 |
| 1  | -2 | 5  |

Transponer Matriz

|    |    |    |
|----|----|----|
| 5  | -2 | 1  |
| -2 | 4  | -2 |
| 1  | -2 | 5  |

$$A^{-1} = \begin{bmatrix} 1 & 5 & -2 & 1 \\ 8 & -2 & 4 & -2 \\ 1 & -2 & 5 & 1 \end{bmatrix} = \begin{bmatrix} \frac{5}{8} & -\frac{1}{4} & \frac{1}{8} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{8} & -\frac{1}{4} & \frac{5}{8} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} \frac{5}{8} & -\frac{1}{4} & \frac{1}{8} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{8} & -\frac{1}{4} & \frac{5}{8} \end{bmatrix} =$$

$$\begin{bmatrix} 2 \cdot \frac{5}{8} + 1 \cdot -\frac{1}{4} + 0 \cdot \frac{1}{8} & 2 \cdot -\frac{1}{4} + 1 \cdot \frac{1}{2} + 0 \cdot -\frac{1}{4} & 2 \cdot \frac{1}{8} + 1 \cdot -\frac{1}{4} + 0 \cdot \frac{5}{8} \\ 1 \cdot \frac{5}{8} + 3 \cdot -\frac{1}{4} + 1 \cdot \frac{1}{8} & 1 \cdot -\frac{1}{4} + 3 \cdot \frac{1}{2} + 1 \cdot -\frac{1}{4} & 1 \cdot \frac{1}{8} + 3 \cdot -\frac{1}{4} + 1 \cdot \frac{5}{8} \\ 0 \cdot \frac{5}{8} + 1 \cdot \frac{1}{2} + 2 \cdot -\frac{1}{4} & 0 \cdot -\frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot -\frac{1}{4} & 0 \cdot \frac{1}{8} + 1 \cdot -\frac{1}{4} + 2 \cdot \frac{5}{8} \end{bmatrix} =$$

$$\begin{bmatrix} \frac{10}{8} - \frac{1}{4} + 0 & -\frac{1}{2} + \frac{1}{2} & \frac{1}{4} - \frac{1}{4} \\ \frac{5}{8} - \frac{3}{4} + \frac{1}{8} & -\frac{1}{4} + \frac{3}{2} - \frac{1}{4} & \frac{1}{8} - \frac{3}{4} + \frac{5}{8} \\ -\frac{1}{4} + \frac{2}{8} & \frac{1}{2} - \frac{2}{4} & \frac{1}{4} + \frac{10}{8} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 5 & -2 & 1 \\ 8 & -2 & 4 & -2 \\ 1 & -2 & 5 & 1 \end{bmatrix} \text{ ou } \begin{bmatrix} \frac{5}{8} & -\frac{1}{4} & \frac{1}{8} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{8} & -\frac{1}{4} & \frac{5}{8} \end{bmatrix}$$

2. Exercícios de aplicação: Sistemas Lineares via Matriz Inversa

Introdução: Em cada exercício

- Escreva o sistema na forma matricial  $AX = B$ ;
- Verifique se a matriz dos coeficientes é invertível;
- Calcule a matriz inversa;
- Determine a solução do sistema utilizando  $X = A^{-1}B$ .

1a. Resolver o sistema:

$$\begin{cases} x + 2y = 4 \\ 3x + 5y = 9 \end{cases}$$

$$\begin{bmatrix} 1 & 2 & 0 & x & = & 4 \\ 3 & 5 & 0 & y & = & 9 \end{bmatrix}$$

$$\begin{aligned} (1 \cdot 5) - (2 \cdot 3) \\ = 5 - 6 \\ = -1 \end{aligned}$$

$$A^{-1} = \frac{1}{-1} \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$$

$$\begin{aligned} x &= -5 \cdot 4 + 2 \cdot 9 \\ &= -20 + 18 \\ &= -2 \end{aligned}$$

$$\begin{aligned} y &= 3 \cdot 4 + (-1) \cdot 9 \\ &= 12 - 9 \\ &= 3 \end{aligned}$$

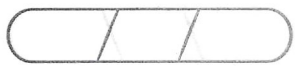
2a. Resolver o sistema:

$$\begin{cases} 2x - y = 4 \\ x + y = 5 \end{cases}$$

$$\begin{bmatrix} 2 & -1 & 0 & x & = & 4 \\ 1 & 1 & 0 & y & = & 5 \end{bmatrix}$$

$$\begin{aligned} (2 \cdot 1) - (-1 \cdot 1) \\ = 2 - (-1) \\ = 3 \end{aligned}$$

$$A^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$



$$x = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} \cdot 5$$

$$y = \frac{-1}{3} + \frac{2}{3} \cdot 5$$

$$x = \frac{1}{3} + \frac{5}{3}$$

$$y = \frac{-1}{3} + \frac{10}{3}$$

$$x = \frac{6}{3}$$

$$y = \frac{9}{3}$$

$$x = 2$$

$$y = 3$$

ic. Resolva o sistema:

$$\begin{cases} x + y + z = 6 \\ 2x + y + 3z = 14 \\ x + 2y + z = 8 \end{cases}$$

|   |   |   |  |   |   |    |
|---|---|---|--|---|---|----|
| 1 | 1 | 1 |  | x |   | 6  |
| 2 | 1 | 3 |  | y | = | 14 |
| 1 | 2 | 1 |  | z |   | 8  |

|   |   |   |   |   |  |
|---|---|---|---|---|--|
| 1 | 1 | 1 | 1 | 1 |  |
| 2 | 1 | 3 | 2 | 1 |  |
| 1 | 2 | 1 | 1 | 2 |  |

1 6 2 1 3 1

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 |
| 2 | 1 | 3 | 0 | 1 | 0 |
| 1 | 2 | 1 | 0 | 0 | 1 |

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 3 - 2 \cdot 1 = 1$$

$$= 0 - 2 \cdot 1 = -2$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 0 - 2 \cdot 0 = 0$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= 2 - 1 = 1$$

$$= 1 - 1 = 0$$

$$= 0 - 1 = -1$$

$$= 0 - 0 = 0$$

$$= 1 - 0 = 1$$

$$\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array}$$

Interca  $L2$  per  $L3$

$$\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & -1 & 1 & -2 & 1 & 0 \end{array}$$

$$L3 \rightarrow L3 + L2$$

$$= -1 + 1 = 0$$

$$= 1 + 0 = 1$$

$$= -2 + (-1) = -3$$

$$= 1 + 0 = 1$$

$$= 0 + 1 = 1$$

$$L1 \rightarrow L1 - L2$$

$$= 1 - 1 = 0$$

$$= 1 - 0 = 1$$

$$= 1 - (-1) = 2$$

$$= 0 - 0 = 0$$

$$= 0 - 1 = -1$$

$$\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & 0 & -1 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -3 & 1 & -1 \end{array}$$

$$L1 \rightarrow L1 - L3$$

$$= 1 - 1 = 0$$

$$= 2 - (-3) = 5$$

$$= 0 - 1 = -1$$

$$= -1 - 1 = -2$$

$$\begin{array}{ccc|ccc} \textcircled{1} & 0 & 0 & 5 & -1 & -2 \\ 0 & \textcircled{1} & 0 & -1 & 0 & 1 \\ 0 & 0 & \textcircled{1} & -3 & 1 & -1 \end{array}$$

$$A^{-1} = \begin{bmatrix} 5 & -1 & -2 \\ -1 & 0 & 1 \\ -3 & 1 & -1 \end{bmatrix}$$

$$\begin{aligned}
 x &= 5 \cdot 6 + (-1) \cdot 14 + 2 \cdot 8 \\
 &= 30 - 14 + 16 \\
 &= 30 - 20 \\
 &= 10
 \end{aligned}$$

$$\begin{aligned}
 y &= -1 \cdot 6 + 0 \cdot 14 + 1 \cdot 8 \\
 &= -6 + 0 + 8 \\
 &= -6 + 8 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 z &= -3 \cdot 6 + 1 \cdot 14 + 1 \cdot 8 \\
 &= -18 + 14 + 8 \\
 &= -18 + 22 \\
 &= 4
 \end{aligned}$$

id. Resolva o sistema:

$$\begin{cases}
 2x + y + z = 5 \\
 x + 2y + z = 6 \\
 x + y + 2z = 9
 \end{cases}$$

|   |   |   |  |   |   |   |
|---|---|---|--|---|---|---|
| 2 | 1 | 1 |  | x | = | 5 |
| 1 | 2 | 1 |  | y | = | 6 |
| 1 | 1 | 2 |  | z | = | 9 |

|   |   |   |   |   |  |                     |
|---|---|---|---|---|--|---------------------|
| 2 | 1 | 1 | 2 | 1 |  | $(8+1+1) - (2+2+2)$ |
| 1 | 2 | 1 | 1 | 2 |  | $= 10 - 6$          |
| 1 | 1 | 2 | 1 | 1 |  | $= 4$               |

$\swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad \swarrow$   
 $8 \quad 1 \quad 1 \quad 8 \quad 1 \quad 1$

|   |   |   |   |   |   |                    |   |   |   |   |   |   |
|---|---|---|---|---|---|--------------------|---|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 0 | 0 | Subtrair L1 por L2 | 1 | 2 | 1 | 0 | 1 | 0 |
| 1 | 2 | 1 | 0 | 1 | 0 |                    | 2 | 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 2 | 0 | 0 | 1 |                    | 1 | 1 | 2 | 0 | 0 | 1 |

L2 -> L2 - 2L1

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 2 = -3$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 0 = 1$$

$$= 0 - 2 \cdot 0 = 0$$

L3 -> L3 - L1

$$= 1 - 1 = 0$$

$$= 1 - 2 = -1$$

$$= 2 - 1 = 1$$

$$= 0 - 0 = 0$$

$$= 0 - 1 = -1$$

$$= -1 - 0 = -1$$

|   |    |    |   |    |   |
|---|----|----|---|----|---|
| 1 | 2  | 1  | 0 | 1  | 0 |
| 0 | -3 | -1 | 1 | -2 | 0 |
| 0 | -1 | 1  | 0 | -1 | 1 |

|                  |   |    |    |   |    |   |
|------------------|---|----|----|---|----|---|
| Subcar L2 per L3 | 1 | 2  | 1  | 0 | 1  | 0 |
|                  | 0 | -1 | 1  | 0 | -1 | 1 |
|                  | 0 | -3 | -1 | 1 | -2 | 0 |

$$L2 \rightarrow -1 \cdot L2$$

$$= -1 \cdot -1 = 1$$

$$= -1 \cdot 1 = -1$$

$$= -1 \cdot 0 = 0$$

$$= -1 \cdot -1 = 1$$

$$= -1 \cdot 1 = -1$$

|   |    |    |   |    |    |
|---|----|----|---|----|----|
| 1 | 2  | 1  | 0 | 1  | 0  |
| 0 | 1  | -1 | 0 | 2  | -1 |
| 0 | -3 | -1 | 1 | -2 | 0  |

$$L1 \rightarrow L1 - 2L2$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot -1 = 3$$

$$= 0 - 2 \cdot 0 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 0 - 2 \cdot 1 = -2$$

$$L3 \rightarrow L3 + 3L2$$

$$= -3 + 3 \cdot 1 = 0$$

$$= -1 + 3 \cdot (-1) = -4$$

$$= 1 + 3 \cdot 0 = 1$$

$$= -2 + 3 \cdot 1 = 1$$

$$= 0 + 3 \cdot 1 = 3$$

|   |   |    |   |    |    |
|---|---|----|---|----|----|
| 1 | 0 | 3  | 0 | -1 | 2  |
| 0 | 1 | -1 | 0 | 2  | -1 |
| 0 | 0 | -4 | 1 | 1  | -3 |

$$L3 \rightarrow -1 \cdot L3$$

$$4$$

$$= -\frac{1}{4} \cdot -4 = \frac{4}{4} = 1$$

$$= -\frac{1}{4} \cdot 1 = -\frac{1}{4}$$

$$= -\frac{1}{4} \cdot 1 = -\frac{1}{4}$$

$$= -\frac{1}{4} \cdot -3 = \frac{3}{4}$$

|   |   |    |                |                |               |
|---|---|----|----------------|----------------|---------------|
| 1 | 0 | 3  | 0              | -1             | 2             |
| 0 | 1 | -1 | 0              | 1              | -1            |
| 0 | 0 | 1  | $-\frac{1}{4}$ | $-\frac{1}{4}$ | $\frac{3}{4}$ |

$$L1 \rightarrow L1 - 3L3$$

$$= 3 - 3 \cdot 1 = 0$$

$$= 0 - 3 \cdot -\frac{1}{4} = \frac{3}{4}$$

$$= -1 - 3 \cdot -\frac{1}{4} = -\frac{1}{4}$$

$$= 0 - 3 \cdot \frac{3}{4} = -\frac{9}{4}$$

$$L2 \rightarrow L2 + L3$$

$$= -1 + 1 = 0$$

$$= 0 + (-\frac{1}{4}) = -\frac{1}{4}$$

$$= 1 + (-\frac{1}{4}) = \frac{3}{4}$$

$$= -1 + \frac{3}{4} = -\frac{1}{4}$$

|   |   |   |                |                |                |
|---|---|---|----------------|----------------|----------------|
| 1 | 0 | 0 | $\frac{3}{4}$  | $-\frac{1}{4}$ | $-\frac{1}{4}$ |
| 0 | 1 | 0 | $-\frac{1}{4}$ | $\frac{3}{4}$  | $-\frac{1}{4}$ |
| 0 | 0 | 1 | $-\frac{1}{4}$ | $-\frac{1}{4}$ | $\frac{3}{4}$  |

$$A^{-1} = \frac{1}{4} \begin{vmatrix} 3 & -1 & -1 \\ 4 & -1 & 3 \\ -1 & -1 & 3 \end{vmatrix}$$

$$x = \frac{1}{4} (3 \cdot 5 + (-1) \cdot 6 + (-1) \cdot 9)$$

$$= \frac{1}{4} (15 - 6 - 9)$$

$$= \frac{1}{4} \cdot 2$$

$$= \frac{1}{2}$$

$$= 2 : 2$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$y = \frac{1}{4} (-1 \cdot 5 + 3 \cdot 6 + (-1) \cdot 9)$$

$$= \frac{1}{4} (-5 + 18 - 9)$$

$$= \frac{1}{4} \cdot 6$$

$$= \frac{1}{2}$$

$$= 6 : 2$$

$$= 3$$

$$= 3$$

$$= 3$$

$$= 3$$

$$z = \frac{1}{4} (-1 \cdot 9 + (-1) \cdot 6 + 3 \cdot 9)$$

$$= \frac{1}{4} (-9 - 6 + 27)$$

$$= \frac{1}{4} \cdot 10$$

$$= \frac{1}{4} \cdot 10$$

$$= 10 : 2$$

$$= 5$$

$$= 5$$

16. Verifique se o sistema admite solução única. Caso admira, resolva

$$\begin{cases} x + 2y + 3z = 6 \\ 2x + 4y + 6z = 12 \\ 3x + y + z = 4 \end{cases}$$

$$2x + 4y + 6z = 12$$

$$3x + y + z = 4$$

tilibra

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| 1 | 2 | 3 |   | x |   | 6  |
| 2 | 4 | 6 | . | y | = | 12 |
| 4 | 4 | 4 |   | y |   | 4  |

|   |   |   |   |   |                                               |
|---|---|---|---|---|-----------------------------------------------|
| 1 | 2 | 3 | 1 | 2 | $(1+12+6) - (12+6+4)$<br>$= 22 - 22$<br>$= 0$ |
| 2 | 4 | 6 | 2 | 4 |                                               |
| 4 | 4 | 4 | 4 | 4 |                                               |

$\swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad \swarrow$   
 $12 \quad 6 \quad 4 \quad 4 \quad 12 \quad 6$

Como  $\det A = 0$ , a matriz é não invertível.